

eVAE: Evolutionary Variational Autoencoder

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Abstract—Variational autoencoders (VAEs) are challenged by the imbalance between representation inference and task fitting caused by surrogate loss. To address this issue, existing methods adjust their balance by directly tuning their coefficients. However, these methods suffer from a tradeoff uncertainty, i.e., nondynamic regulation over iterations and inflexible hyperparameters for learning tasks. Accordingly, we make the first attempt to introduce an *evolutionary* VAE (eVAE), building on the variational information bottleneck (VIB) theory and integrative evolutionary neural learning. eVAE integrates a variational genetic algorithm (VGA) into VAE with variational evolutionary operators, including variational mutation (V-mutation), crossover, and evolution. Its training mechanism synergistically and dynamically addresses and updates the learning tradeoff uncertainty in the evidence lower bound (ELBO) without additional constraints and hyperparameter tuning. Furthermore, eVAE presents an evolutionary paradigm to tune critical factors of VAEs and addresses the premature convergence and random search problem in integrating evolutionary optimization into deep learning. Experiments show that eVAE addresses the KL-vanishing problem for text generation with low reconstruction loss, generates all the disentangled factors with sharp images, and improves image generation quality. eVAE achieves better disentanglement, generation performance, and generation-inference balance than its competitors. Code available at: <https://github.com/amasawa/eVAE>.

Index Terms—Evolutionary variational autoencoder (eVAE), variational autoencoder (VAE), variational genetic algorithm (VGA).

I. INTRODUCTION

VARIATIONAL autoencoders (VAEs) [24] have attracted significant interest for their capability of learning continuous and smooth distributions from observations by integrating probabilistic and deep neural learning principles. VAEs have demonstrated significant advantages of incorporating prior knowledge, mapping inputs to probabilistic representations, and approximating the likelihood of outputs. Their integration of a stochastic gradient variational Bayes (SGVB) estimator [24] with neural settings learns a narrow probabilistic latent space to infer more representative attributes in a hidden space. VAEs have been applied in various domains,

including time series forecasting [13], out-of-domain detection in images [17], [30], [31], [39], generating images with spiking signals [21], and generating text by language modeling [42]. Beyond generative tasks, VAEs are widely used in representation learning tasks, particularly for disentanglement [33], [41], classification [20], [35], clustering [40], and manifold learning [1], [9]. However, VAEs still face significant issues in particular learning an appropriate tradeoff between representation compression and generation fitting.

Theoretically, the bound optimization of variational inference is introduced to substitute the log-likelihood function with a surrogate function to optimize it by gradient descent. In practice, the evidence lower bound (ELBO) cannot approach the maximum of conditional likelihood with a close gap between posterior and prior. It fails to achieve a tradeoff between representation robustness and reconstruction effectiveness. Specifically, a weak KL divergence could incur KL vanishing, while a strong KL divergence could result in a bad likelihood. Furthermore, the tradeoff is sensitive to posterior distribution disjoint, data characteristics, and network architectures [5].

Recently, various techniques have been proposed to address these issues. The first set aims to adjust the term balance in objective functions. Examples are β -VAEs [6], [18] incorporating a hyperparameter β , InfoVAE [43] adding a scaling parameter to the KL term, SA-VAE [14] having a cyclical annealing schedule to progressively increase β for reducing KL vanishing, and ControlVAE [34] introducing the proportional-integral-derivative (PID) control to tune hyperparameter. However, these are partial solutions only adjusting one part of the objectives, failing to weigh and resolve the balance issue in dynamic settings.

From the information bottleneck (IB) perspective, VAE can be treated as a lossy information compressor. Adjusting the KL within a range controls the IB flowing from representing latent variables to reconstructing samples. It thus benefits the tradeoff between compression and reconstruction [3], [6]. This variational IB (VIB) inspires us to dynamically search the optimal state of the compressor in iterations to fit information flow. To capture the evolving factor β in iterations and the optimal bottleneck embedded by a fitness function, we introduce a novel *evolutionary* VAE (eVAE). eVAE incorporates and integrates variational evolutionary learning and VIB into VAE. It achieves a better optimum exploration and an appropriate tradeoff between representation compression and generation fitting.

Integrating evolutionary learning into VAE is an open topic. We make the first attempt by proposing a *variational genetic algorithm* (VGA) to optimize VAE objectives and its

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modeling exploration in an evolving manner. Furthermore, eVAE dynamically optimizes the VAE inference (the KL terms) in an evolutionary and probabilistic manner, which further induces a stable convergence toward balancing inference and generation. Specifically, to avert premature convergence, eVAE introduces probabilistic chromosome selection for a smooth search space. To avoid exhaustive random search, we apply the simulated binary crossover (SBX) [10] and Cauchy distributional mutation to guide the training toward a stable convergence. In summary, eVAE forms the first evolutionary VAE framework. It does not empower constraints on VAE architectures, input settings, and objective function (ELBO). eVAE integrates variational encoding and decoding, IB, and evolutionary learning into the deep neural framework. The evaluation against the state-of-the-art VAEs shows that eVAE achieves better disentanglement and reconstruction loss. It addresses the KL vanishing effectively.

The main contributions of this work are as follows.

- 1) We propose the eVAE model to achieve a tradeoff between inference and generation. Specifically, a Cauchy-based variational mutation (V-mutation) operator and an SBX based variational crossover (V-crossover) operator train a model with an evolving inner–outer joint training algorithm. It tackles the premature convergence and random search problem when optimizing deep models, assisted by a VGA.
- 2) We illustrate eVAE by deriving the lower bound on β -VAE-related models under the IB theory. We analyze the information flow in the corresponding lower bound per the rate–distortion theory, which shows that only the iteration-specific lower bound in eVAE can capture the optimal trend for effective training.
- 3) We verify the effectiveness of eVAE by applying it to three tasks: disentangled representation learning on dSprites, image generation on CelebA, and text generation on the PTB corpus.

II. VAE BACKGROUND AND ISSUES

Here, we briefly introduce the background of VAEs by focusing on their objectives and mechanisms to address the inference–generation balance. These are mostly relevant to the target in our work. Other developments on VAEs, such as adjusting VAE network structures, input settings, and learning constraints, are excluded due to their irrelevance to this work.

1) *VAEs and Challenges*: VAEs mitigate autoencoder issues such as sparse representation [36] by learning a continuous and smooth representation distribution $p(\mathbf{x}, \mathbf{z})$, $\mathbf{x} \in \mathcal{X}$ from observations $\mathcal{X}$ over latent variables $\mathbf{z}$. After learning the posterior distribution $q_\phi(\mathbf{z} | \mathbf{x})$ for an encoding neural network, a VAE network applies variational inference to approximate the decoder distribution $p_\theta(\mathbf{x} | \mathbf{z})$. Learning tasks such as reconstructed and generated outputs can then be sampled from this learned distribution in a generative process. With the SGVB estimator and reparameterization trick, the gradients become tractable, and the inference parameters ϕ and generative parameters θ are learnable. The objectives of VAEs can be converted into ELBO with the expectation over empirical

distribution p_{data} of the data toward both inference $\mathcal{L}_I$ and reconstruction $\mathcal{L}_R$ [12], [15]

$$\mathcal{L}_{\text{ELBO}} = E_{\mathbf{x} \sim p_{\text{data}}} \left[\underbrace{E_{q_\phi(\mathbf{z} | \mathbf{x})} [\log p_\theta(\mathbf{x} | \mathbf{z})]}_{\mathcal{L}_R} - \underbrace{D_{KL}(q_\phi(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z}))}_{\mathcal{L}_I} \right]. \quad (1)$$

Ideally, the surrogate loss function in (1) should result in both a precise latent space and appropriate task fitting. However, in practice, VAEs often prioritize data fitting over accurate inference. For instance, VAEs with a weak capacity encoder may degenerate into autoregressive models in variational language generation tasks, causing the KL divergence to vanish quickly. On the other hand, a powerful decoder that learns orthogonal transformations of posterior distributions may result in inaccurate inference in representation tasks [23], [29]. Moreover, there exists a training conflict [43] between inference and generation arising from ELBO. Specifically, penalizing $\mathcal{L}_R$ will strengthen the disjointness between $q_\phi(\mathbf{z} | \mathbf{x}^i)$ to separate the log-likelihood concentrated on each sample $\mathbf{x}^i$. As a result, the mass of the joint distribution is maximized. However, this approach conflicts with the joint assumption in the posterior space.

2) *VAE Variants*: One way to address the above VAE issues is to tighten the log-likelihood lower bound for the correct variational approximation of posterior [2], [11], [19], [26], [32], [38] or prior [25], [37] or the appropriate decomposition of ELBO [12], [43] under some mild assumptions. For example, β -VAE [18] adds the hyperparameter β to weigh the $\mathcal{L}_{KL}$ term. Then, ELBO minimizes $\mathcal{L}_R$ to the convergence of data fitting collectively with regularization $\mathcal{L}_I$ by varying β

$$\mathcal{L}_{\text{ELBO}} = E_{\mathbf{x} \sim p_{\text{data}}} [\mathcal{L}_R + \beta \mathcal{L}_I]. \quad (2)$$

β -VAE introduces some fundamental limitations, which trigger various follow-up research. InfoVAE [43] introduces a scaling parameter λ on the KL-term and converts the objective into

$$\mathcal{L}_{\text{ELBO}} = \alpha I_q(\mathbf{x}; \mathbf{z}) - \lambda D_{KL}(q_\phi(\mathbf{z}) \| p(\mathbf{z})) - E_{q(\mathbf{z})} [D_{KL}(q_\phi(\mathbf{x} | \mathbf{z}) \| p_\theta(\mathbf{x} | \mathbf{z}))] \quad (3)$$

where I_q is the mutual information with weights α and $\alpha + \lambda - 1 = 0$.

SA-VAE involves a cyclical annealing schedule to split the training into multiple cycles starting at $\beta = 0$ and progressively increases β until $\beta = 1$ to reduce the KL vanishing [14]. In ControlVAE [34] and DynamicVAE [33], the PID controller compares the KL divergence with a set point, and their difference is used as feedback to the controller to tune hyperparameter $\beta(t)$. ControlVAE and DynamicVAE thus optimize KL dynamically but are constrained by the PID controller, which follows a separate tuning mechanism of VAE.

The existing work leaves gaps for constructing an approximate weight allocation between reconstruction and inference during iterations. Specifically, hyperparameter tuning in an outer loop may be irrelevant to VAE behaviors. Optimization is nondynamic and constrained on data and networks. Heuristic components have difficult-to-tune hyperparameters.

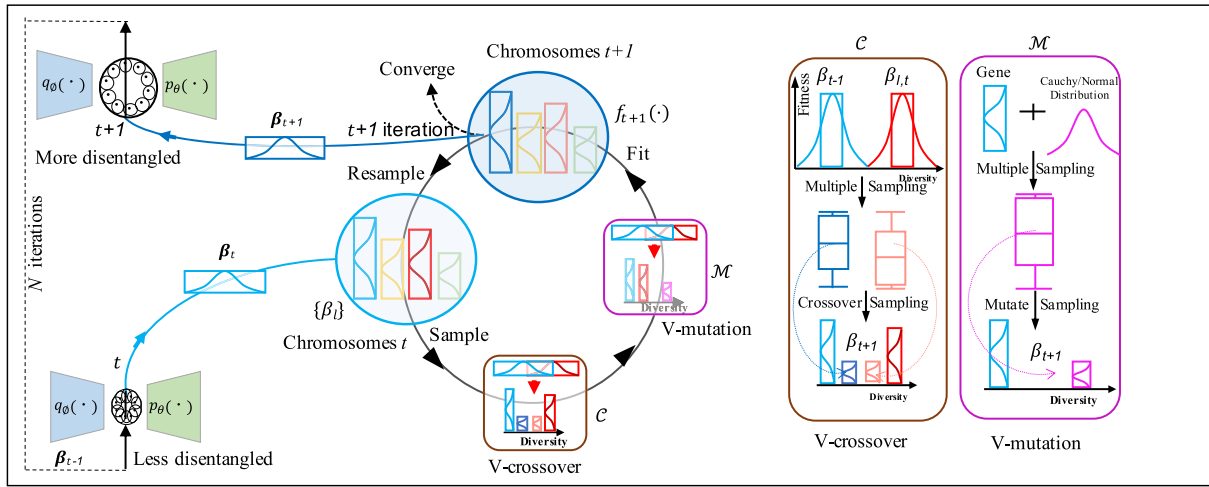


Fig. 1. Framework of eVAE. The VAE results inform chromosome sampling. The genes are then updated by variational V-crossover and V-mutation. The evolved results at t are checked per fitness for retraining, giving up, or convergence at $t + 1$.

3) *VAE Gaps*: To obtain representations for downstream tasks, the existing VAE methods often pursue a reconstruction tradeoff for better inference.

By regularizing the approximated or aggregated posterior distribution [2], [11], [19], [25], [26], [32], [37], [38], the meta-prior-based methods explicitly capture causal dependencies among latent variables. Hence, such explicit relational inductive bias can model the temporal or spatial coherence between features, yielding task-specific representations. However, mismatches between the aggregated meta-priors and optimal priors can disrupt the latent space, causing a representation hole [22] problem. This hampers generative tasks and limits the extensibility of other representation learning tasks.

The methods based on ELBO decomposition [12], [43] typically take the factorization assumption and implicitly learn the intra- and interfeature dependencies using statistical or deep learning techniques. This implicit relational inductive bias models non-IID relationships in latent variables to achieve disentanglement. However, a tighter ELBO can prioritize inference during training while sacrificing reconstruction. Furthermore, statistical methods are typically more suitable for smaller datasets, whereas deep learning techniques may experience posterior collapse because of inconsistent discriminator performance.

Instead of using meta-prior-based approaches, our method avoids making extra prior assumptions. A fitness function in the evolving inner–outer joint training enhances both representation and inference, side-stepping the inaccuracy that comes with fixed priors in amortized inference.

Unlike ELBO-based models, we retain the vanilla VAE learning objective. The evolutionary operators in the evolving inner–outer joint training fine-tune the VAE objective by β in (2), achieving smooth compressed representation without sacrificing decoding quality.

Compared with simple heuristic-based methods [33], [43], our adjustment is dynamic. It not only avoids manual setting errors but also ensures a better compression space guided by the iteration-specific IB lower bound. Qualitative and quantitative experiments show that our approach achieves comparable results across various modalities and learning tasks.

III. EVAE MODEL

eVAE addresses the aforementioned gaps by incorporating novel variational genetic learning into balancing inference and generation and evolutionarily involving their effect in adjusting VAE learning behaviors toward a better tradeoff without complex hyperparameter tuning. eVAE also tunes the inference–generation balance in (1) by jointly addressing the above issues in (2) and (3). Fig. 1 illustrates the eVAE framework of integrating variational evolutionary learning into deep learning to improve the VAE balance.

A. Notation and Problem Setup

We denote the input variable as $\mathbf{x} \in \mathcal{X} \subseteq \mathcal{R}^{d_x}$ and let $\mathbf{z} \in \mathcal{Z} \subseteq \mathcal{R}^{d_z}$ be a latent variable ($d_x \gg d_z$). To achieve optimal inference capability, i.e., $\mathbf{z}$ for downstream tasks, and superior reconstruction ability, i.e., reconstructed $\hat{\mathbf{x}}$ generated by the decoder from $\mathbf{x}$, the VAE optimizes a surrogate loss in (2) regulated by a factor β during each iteration. We adopt an evolutionary algorithm-based training method, i.e., *evolving inner-outer joint training*, where an appropriate value of β is determined to balance the tradeoff between the model's inference and reconstruction, and this process evolves at each iteration.

The learning framework we are examining is structured as follows: at the start of training, a set of candidate values, $\{\beta\}$, containing l chromosomes, are produced. For the t th iteration, the model evaluates its distortion $\mathcal{L}_I$ and rate $\mathcal{L}_R$. These metrics are subsequently input to the variational evolutionary learner $\mathcal{E}$. Operated under the probabilities of Pr_c and Pr_m , this learner carries out crossover and mutation operations on the chromosomes within the population. These adjusted genes are then integrated into the model for the $t + 1$ th iteration as depicted in (2). This methodology ensures the model navigates a trajectory that adeptly balances inference and reconstruction.

B. eVAE—Evolving Inner–Outer Joint Training

The eVAE implements an evolving learning process of optimizing both VAE and its evolutionary parameters through an inner–outer joint iterative training process, as shown in

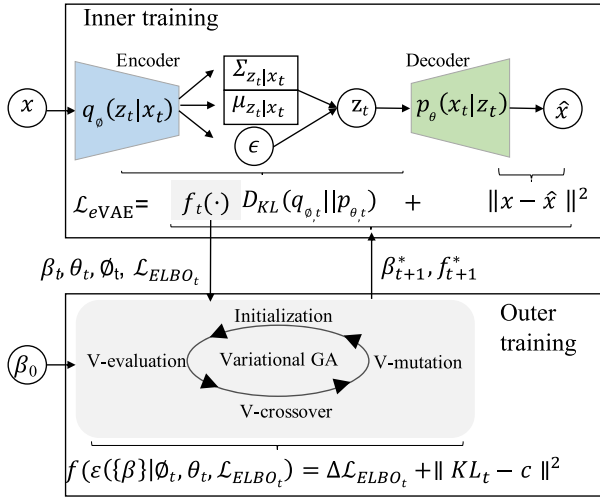


Fig. 2. eVAE—inner-outer joint evolutionary training process. The upper part shows the VAE training at time t . The objectives are then incorporated into the lower part—the outer training by VGA, whose fitness-based optimized results are fed to the VAE for further training.

Fig. 2. At time t , given input $\mathbf{x}_t \in \mathcal{X}$, a VAE inner training process trains a VAE model $\mathcal{V}$ (we do not constrain the VAE framework, for generality, we use β -VAE for case study) with some initialization β_0 to learn the decoder $p_\theta(\mathbf{x}_t | \mathbf{z}_t)$ and the encoder $q_\phi(\mathbf{z}_t | \mathbf{x}_t)$ with prior $p(\mathbf{z})$. We obtain the VAE inner objective function with initial weight β_0

$$\mathcal{L}_{\text{ELBO}_t^{\text{inner}}} = E_{\mathbf{x}_t \sim p_{\mathcal{X}}} [\mathcal{L}_R(\mathbf{x}_t, \mathbf{z}_t) + \beta \mathcal{L}_I(\mathbf{x}_t, \mathbf{z}_t)]. \quad (4)$$

Then, we introduce an outer variational evolutionary learner $\mathcal{E}$ to update the initial weight in an outer process. In this work, we instantiate $\mathcal{E}$ by a VGA for variational evolution. $\mathcal{E}$ with the fitness function f samples a chromosome β_t from an evolving distribution $\beta_t \sim \mathcal{R}$ and evolves it by taking V-crossover and mutation to generate a new individual β_{t+1} for next outer iteration.

The evolved parameter is then incorporated into $\mathcal{V}_t$ for the next iteration $t+1$ training. We save the states of VAEs at time t and generate the fitness value of β_{t+1} guided by (15). Consequently, (4) is evolved to

$$\mathcal{L}_{\text{ELBO}_{t+1}^{\text{outer}}} = E_{\mathbf{x}_{t+1} \sim p_{\mathcal{X}}} [\mathcal{L}_R(\mathbf{x}_{t+1}, \mathbf{z}_{t+1}) + \beta \mathcal{L}_I(\mathbf{x}_{t+1}, \mathbf{z}_{t+1})]. \quad (5)$$

Furthermore, we reload the state of VAEs at time t and update β_{t+1} in inner training iterations by the optimal β_{t+1}^* , corresponding to the β value in its candidate group $\{\beta\}$ with the highest fitness value over f . Then, this results in updated $\mathcal{V}_{t+1}$ and its parameters ϕ_{t+1} , θ_{t+1} , and β_{t+1} for the $t+1$ inner iteration. We further repeat VGA $\mathcal{E}$ to obtain another set of parameters and retrain model $\mathcal{V}_{t+1}$ until it converges.

The above inner-outer iterative training progresses over time and samples in a generative process to iteratively optimize the balance between reconstruction (i.e., minimizing $\mathcal{L}_R = \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|$ with the reconstructed $\hat{\mathbf{x}}_t$) and inference (i.e., seeking appropriate $\mathcal{L}_I$) and the VAE learning behaviors. In this way, VAE $\mathcal{V}$ is optimized until it converges.

Accordingly, the eVAE generative process optimizes the following objective function:

$$\mathcal{L}_{\text{eVAE}} = \min_{\theta_t, \phi_t} \sum_{t=1}^N \mathcal{L}_{\text{ELBO}_{t+1}}(\theta_t, \phi_t; f(\mathcal{E}(\phi_t, \theta_t, \{\beta\}, \mathcal{L}_{\text{ELBO}_t})), \mathbf{x}_t). \quad (6)$$

To approach β^* for the optimal balance between representation and construction

$$\beta^* \sim f(\mathcal{E}(\{\beta\} | \phi_t, \theta_t, \mathcal{L}_{\text{ELBO}_t})). \quad (7)$$

The overall ELBO of eVAE can then be represented by

$$\begin{aligned} \mathcal{L}_{\text{eVAE}} &= E_{\mathbf{z} \sim q_\phi(\mathbf{z}_t | \mathbf{x}_t)} \log p_\theta(\mathbf{x}_t | \mathbf{z}_t) \\ &\quad + f_{\beta_t \sim \mathcal{R}}(\beta_t, \mathcal{E}) D_{KL}(q_\phi(\mathbf{z}_t | \mathbf{x}_t) \| p(\mathbf{z})) \\ &= E_{\mathbf{x}_{t+1} \sim p_{\mathcal{X}}} [\mathcal{L}_R + f(\beta) \mathcal{L}_I + \mathcal{E}(\Delta \mathcal{L}_{\text{ELBO}})]. \end{aligned} \quad (8)$$

C. Variational Evolution in eVAE

Here, we introduce the *variational evolutionary learner* $\mathcal{E}$ to optimize the parameters. We instantiate $\mathcal{E}$ by the VGA with V-crossover and mutation operations. This avoids typical issues, including premature convergence and random search during discordant exploitation and exploration.

1) *Variational Genetic Algorithm*: The eVAE integrates the VGA-based outer parameter optimization with the VAE-internal gradient-descent-based optimization. Accordingly, VGA consists of a few variational steps and operations: initialization, V-crossover, V-mutation, and variational evaluation (V-evaluation), as shown in the bottom part of Fig. 2. Below, we introduce them, respectively.

a) *Chromosome selection*: To enable a larger and smoother GA search space compatible with VAEs, chromosomes are embedded by a continuous variable β , sampled from an evolving distribution $\mathcal{R}$ (e.g., in the crossover). Assume L individual chromosomes form a candidate group $\{\beta_l\} = \{\beta_1, \dots, \beta_L\}$, a chromosome is chosen from this candidate group to train the VAE, which is associated with a fitness value f after VGA operations and VAE retraining. We thus have the chromosome-fitness pairs embedded for VGA evolution and offspring selection over generations, such as

$$\{\beta, f\} = \{(\beta_1, f_1), \dots, (\beta_L, f_L)\} \quad (9)$$

to optimize the inner-outer joint training.

b) *V-crossover*: Paired strong chromosomes at time t are crossed over to generate new genes for improving the genetic variety of offspring. Following the SBX method [10], we implement the following crossover operations $\mathcal{C}$. It induces a large search space evolving over the strongest chromosome $\beta_{t,\text{father}}$ with the highest fitness value at time t , and the second strongest chromosome $\beta_{t,\text{mother}}$ at time t selected from the chromosome candidate group $\{\beta_l\}$. This obtains the offspring candidate chromosome β_{t+1} for the next time $t+1$ in strategy $\mathcal{C}$

$$\mathcal{C}: \begin{cases} \beta_{t+1,\text{child1}} = \frac{1}{2}[(1+r_c)\beta_{t,\text{father}} + (1-r_c)\beta_{t,\text{mother}}] \\ \text{or} \\ \beta_{t+1,\text{child2}} = \frac{1}{2}[(1-r_c)\beta_{t,\text{father}} + (1+r_c)\beta_{t,\text{mother}}] \end{cases} \quad (10)$$

where r_c is the crossover rate, which is drawn from a probability density function $P_c(r_c)$

$$P_c(r_c) = \begin{cases} 0.5(\eta + 1)r_c^\eta, & \text{if } r_c \leq 1 \\ 0.5(\eta + 1)\frac{1}{r_c^{\eta+2}}, & \text{otherwise} \end{cases} \quad (11)$$

by concentrating on their corresponding parents $\beta_{t,\text{father}}$ and $\beta_{t,\text{mother}}$. By sampling r_c from the following function:

$$r_c = \begin{cases} (2u)^{\frac{1}{\eta+1}}, & \text{if } u \leq 0.5 \\ \left(\frac{1}{2(1-u)}\right)^{\frac{1}{\eta+1}}, & \text{otherwise} \end{cases} \quad (12)$$

where u is a random variable, and η is a hyperparameter for scaling. The parameter η governs the similarity between the offspring produced and their parents. A larger value of η results in offspring closer to their parents, while a smaller value leads to offspring with greater variations. The offspring β_{t+1} can then be created by selecting the better $\mathcal{C}$ strategy [i.e., better satisfying (15)].

The whole process is illustrated in Algorithm 1.

Algorithm 1 VGACrossover: Simulated Binary Crossover in Variational Genetic Algorithm

```

1: Input: The first and second strongest chromosomes at  $t$ -iteration  $\beta_{t,\text{father}}, \beta_{t,\text{mother}}$ , hyperparameter for scaling  $\eta$ 
2: Output:  $\{\beta_l\}^{t+1}$  // group at  $t+1$ -iteration
3:   Generate a random variable  $u$  between 0 and 1;
4:   if  $u \leq 0.5$  then
5:      $r_c = (2u)^{\frac{1}{\eta+1}}$ ; //Generate a mutation rate  $r_c$ .
6:   else
7:      $r_c = (\frac{1}{2(1-u)})^{\frac{1}{\eta+1}}$ ;
8:   end if
9:   function SBXCROSSOVER( $\beta_{t,\text{father}}, \beta_{t,\text{mother}}, r_c$ );
10:  //Generate offspring based on  $\beta_{t,\text{father}}, \beta_{t,\text{mother}}, r_c$ ;
11:   $\beta_{t+1,\text{child1}} = 0.5[(1+r_c)\beta_{t,\text{father}} + (1-r_c)\beta_{t,\text{mother}}]$ ;
12:   $\beta_{t+1,\text{child2}} = 0.5[(1-r_c)\beta_{t,\text{father}} + (1+r_c)\beta_{t,\text{mother}}]$ ;
13:  // Update candidate group  $\{\beta_l\}^t$ 
14:  Choose the two chromosomes with the lowest fitness value from group  $\{\beta_l\}^t$ , i.e.,  $\beta_{t,w}, \beta_{t,sw}$ ;
15:   $\beta_{t+1,w} = \beta_{t+1,\text{child1}}$ ;
16:   $\beta_{t+1,sw} = \beta_{t+1,\text{child2}}$ ;
17: end function
18: Return:  $\{\beta_l\}^{t+1}$ 

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c) *V-mutation*: The crossover-generated offspring are further mutated to improve their genomic diversity and evolutionary capacity. Aligning with V-crossover, a V-mutation strategy $\mathcal{M}$ is taken to diversify the offspring to update β_{t+1} inherited from the crossover or β_t from the current moment. With a relatively small probability, a chromosome $\beta_{t,m}$ selected from the group is mutated as

$$\mathcal{M} : \beta_{t+1} = \beta_{t,m} + r_m \quad (13)$$

where r_m is a random variable sampled from the Cauchy distribution P_m

$$P_m = \frac{1}{\pi} \left(\frac{1}{1 + \beta_{t,m}^2} \right). \quad (14)$$

Refer to Algorithm 2 for the detailed procedure.

Algorithm 2 VGAMutation: Cauchy-Based Mutation in Variational Genetic Algorithm

```

1: Input:  $\beta_{t,m}$  //The randomly selected chromosomes at  $t$ -iteration picked from group  $\{\beta_l\}^t$ 
2: Output:  $\{\beta_l\}^{t+1}$  // group at  $t+1$ -iteration, i.e.,  $\beta_{t+1}$ 
3: function CAUCHYMUTATION( $\beta_{t,m}$ )
4:   Generate a Cauchy distribution based on  $\beta_{t,m}$ ;
5:    $P_m = \frac{1}{\pi} \frac{1}{1 + \beta_{t,m}^2}$ ;
6:   Sampling from  $P_m$ ;
7:   //Generate a mutated chromosome
8:    $\beta_{t+1} = \beta_{t,m} + r_m$ ; //update the group
9: end function
10: Return:  $\{\beta_l\}^{t+1}$ 

```

2) *V-Evaluation and VGA Fitness Function*: The chromosomes updated by the V-crossover and V-mutation operations are evaluated in alignment with the VAE objectives to determine whether they are transferable to the next generation. Within the VAE framework, we take the following heuristic fitness function f to guide the evolution of chromosomes and their fitness to VAE objectives. The fitness function integrates the direction of VAE-oriented stochastic gradient descent and the distance to the optimal KL divergence to achieve multiobjectives

$$f_{t+1} = \Delta \mathcal{L}_{\text{ELBO}_{t+1}} + \|KL_{t+1}(\beta_{t+1}) - c\| \quad (15)$$

where $\Delta \mathcal{L}_{\text{ELBO}_{t+1}}$ is

$$\Delta \mathcal{L}_{\text{ELBO}_{t+1}} = \mathcal{L}_{\text{ELBO}_{t+1}}(\beta_{t+1}) - \mathcal{L}_{\text{ELBO}_t}(\beta_t) \quad (16)$$

denoting the ELBO of VAE difference after applying the evolved β . c is the task-specific IB, which ensures the bound optimization of eVAE over the evolutionary optimization.

V-evaluation and the fitness f guide eVAE to converge and balance the reconstruction and inference with enhanced evolutionary parameterization. The larger the fitness value, the stronger the chromosome in its candidate group, which forms a strong pair $\{\beta_{t+1}^*, f_{t+1}^*\}$.

Algorithm 3 illustrates the modeling process of eVAE. It consists of an inner–outer joint training of VAE and evolutionary learning iteratively.

IV. THEORETICAL ANALYSIS

We illustrate the eVAE framework by taking β -VAE as a base model to build an evolutionary VAE. Below, we analyze the eVAE effect of adjusting parameters, the tradeoff between reconstruction and regularization, and training performance in β -VAE.

Algorithm 3 EIOTraining: Evolving Inner–Outer Joint Training

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1: Input: crossover rate  $Pr_c$ , mutation rate  $Pr_m$ , batch data  $\mathbf{x}$ .
2: Output:  $\{\beta^*, f^*\}$ 
3: Initialization: parameters of decoder  $\theta$ , parameters of encoder  $\phi$ ,  $\{\beta_l\} = \{\beta_1, \dots, \beta_L\}$  //initialize group
4: while  $t < T$  do
5:    $Pr_t \leftarrow$  sample from  $N(0, 1)$ ; //probability to evolve
6:   if  $Pr_t \leq Pr_m$  then
7:      $\theta_t, \phi_t$ ; //save current variational parameters
8:     generate  $\beta_{t+1}$  by V-mutation  $\mathcal{M}(\beta_t)$  per (13) and (14);
9:     evaluate  $\beta_{t+1}$  per fitness function  $f(\mathcal{E}(\phi_t, \theta_t, \{\beta\}, \mathcal{L}_{\text{ELBO}}))$  by (15) and (16);
10:  else if  $Pr_m < Pr_t \leq Pr_c$  then
11:     $\theta_t, \phi_t$ ; //save current variational parameters
12:    generate  $\beta_{t+1}$  by V-crossover  $\mathcal{C}(\beta_{t,\text{father}}, \beta_{t,\text{mother}})$  per (10), (11) and (12);
13:    evaluate  $\beta_{t+1}$  by  $f(\mathcal{E}(\phi_t, \theta_t, \{\beta\}, \mathcal{L}_{\text{ELBO}}))$  per (15) and (16);
14:  else
15:     $\{\beta^*, f^*\} \leftarrow \beta_t$ ; //select the strongest pair
16:     $\theta_{t+1}, \phi_{t+1} \leftarrow \mathcal{L}_{\text{eVAE}}(\theta_t, \phi_t, \beta^*|\mathbf{x}_t)$ ; // update variational parameters
17:  end if
18:  Return:  $\{\beta^*, f^*\}$ 
19: end while

```

To elucidate the effective compression performance of encoders and decoders in training eVAE, we revisit ELBO in VAEs from a VIB perspective. Guided by the rate–distortion theory, we define an optimization bound for β -VAE. Accordingly, we can categorize the experiment-specific lower bound in β -VAE, the KL-specific lower bound in controlVAE, and the iteration-specific lower bound in eVAE. The first type [6], [18] refers to predefined methods, where the ratio β_B between the reconstruction and representation functions is tuned as a hyperparameter. The second category is the PID control-based methods [34]. They use a set of nonlinear controllers of KL divergence to manipulate the coefficient β_{KL} explicitly. Finally, eVAE aims to dynamically tune the coefficient by a variational evolutionary learner $\mathcal{E}(\beta_t)$ in iterations. Furthermore, with the rate–distortion ($R - D$) curve in three models over iterations, we analyze the effect of eVAE in adjusting β -VAE parameters and its tradeoff between reconstruction and regularization and training performance theoretically and experimentally below.

Specifically, β -VAE directly tunes hyperparameter β by setting a task-relevant ratio between the inference loss $\mathcal{L}_I$ and the generation loss $\mathcal{L}_R$. In essence, incrementally regularizing $\mathcal{L}_I$ by β is equivalent to maximizing the likelihood on an

experiment-specific threshold, i.e., a constant B [18]

$$\max_{\phi, \theta} E_{\mathbf{x} \sim p_{\text{data}}} \mathcal{L}_R, \quad \text{s.t. } \beta \mathcal{L}_I < B. \quad (17)$$

This learning process can be interpreted in terms of the IB theory [2]. The $\beta \mathcal{L}_I$ constraint operates as the bottleneck, compressing the capacity of latent variable $\mathbf{Z}$ organized from input $\mathbf{X}$ to expressively represent the target $\hat{\mathbf{X}}$ [3]. Accordingly, IB maximizes $I(\mathbf{Z}, \hat{\mathbf{X}})$ to acquire a concise representation

$$\max I(\mathbf{Z}, \hat{\mathbf{X}}), \quad \text{s.t. } I(\mathbf{Z}, \hat{\mathbf{X}}) \leq I_B. \quad (18)$$

Theorem 1: For the hyperparameter β_B in β -VAE, we have the experiment-specific lower bound

$$\mathcal{L}_{\beta\text{-VAE}}(\theta, \phi) = -D - \beta_B R \leq -\beta_B I - D$$

where R refers to the compression rate, D refers to the distortion measuring the representation relevance to a task, and β_B is an experiment-specific constant. VAEs can thus be viewed as a process of achieving the rate–distortion tradeoff.

Proof: According to the VIB theory, ELBO can be bounded by distortion $D = -E_{q_\phi(\mathbf{x}, \mathbf{z})}[\log p_\theta(\mathbf{x} | \mathbf{z})]$ and rate $R = E_{q_\phi(\mathbf{x}, \mathbf{z})}[D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))]$, where $I(\mathbf{X}, \mathbf{Z})$ has the upper bound $I \leq R$

$$\begin{aligned} R &= \int q(\mathbf{x}) d\mathbf{x} \int q_\theta(\mathbf{z} | \mathbf{x}) \log \frac{q_\theta(\mathbf{z} | \mathbf{x})}{p(\mathbf{z})} d\mathbf{z} \\ &= I(\mathbf{X}, \mathbf{Z}) + TC(\mathbf{Z}) \\ &\geq I(\mathbf{X}, \mathbf{Z}) \end{aligned}$$

and the lower bound $I \geq H - D$

$$\begin{aligned} I &= \int \int p(\mathbf{x}, \mathbf{z}) \log \frac{p(\mathbf{x} | \mathbf{z})}{p(\mathbf{x})} d\mathbf{x} d\mathbf{z} \\ &\geq \int \int p(\mathbf{x}, \mathbf{z}) \log \frac{q(\mathbf{x} | \mathbf{z})}{p(\mathbf{x})} d\mathbf{x} d\mathbf{z} \\ &= \int \int p(\mathbf{x}, \mathbf{z}) \log q(\mathbf{x} | \mathbf{z}) d\mathbf{x} d\mathbf{z} - \int p(\mathbf{x}) \log p(\mathbf{x}) d\mathbf{x} \\ &= \int \int p(\mathbf{x}, \mathbf{z}) \log q(\mathbf{x} | \mathbf{z}) d\mathbf{x} d\mathbf{z} + H(\mathbf{X}) \\ &= H - D. \end{aligned}$$

All in all, the VIB can be regarded as

$$H - D \leq I \leq R$$

where H is the entropy of the data.

In the β -VAE framework, β_B imposes an experiment-specific constraint on the compressor and decompressor, producing the variational lower bound

$$\begin{aligned} H - D &\leq I \leq R \\ H - D &\leq \beta_B I \leq \beta_B R \\ H &\leq \beta_B I + D \leq \beta_B R + D. \end{aligned}$$

In summary, the lower bound can be denoted as

$$\mathcal{L}_{\beta\text{-VAE}}(\theta, \phi) = -D - \beta_B R \leq -\beta_B I - D. \quad \blacksquare$$

ControlVAE is a variant of β -VAE, tuning the KL-specific β_{KL} with a PID controller to a desired KL point.

Theorem 2: For the KL-specific factor β_{KL} in ControlVAE, we have

$$\mathcal{L}_{\text{ControlVAE}}(\theta, \phi) = -D - \beta_{KL}R \leq -\beta_{KL}I - D.$$

The eVAE extends the β -VAE framework by tuning the IB $\beta I(\mathbf{X}, \mathbf{Z})$ through a variational evolutionary learner $\mathcal{E}$. After iteration t of the outer training, β_t evolves to fit the task and training phase, guided by the fitness function f .

Theorem 3: The eVAE generates a lower bound to optimize the inner training phase at iteration $t + 1$

$$\mathcal{L}_{\text{eVAE}} \leq -\mathcal{E}(\beta_t)I_t - \mathcal{E}(\beta_t)D_t$$

where an evolving iteration-specific lower bound can strike a compromise between task fitting in $-\mathcal{E}(\beta_t)D_t$ and inference quality in $\mathcal{E}(\beta_t)I_t$.

Proof: According to (8) and (15), we can obtain the loss function of eVAE at iteration t as follows:

$$\begin{aligned} \mathcal{L}_{\text{eVAE}} &= E_{\mathbf{z} \sim q_\phi(\mathbf{z}_t | \mathbf{x}_t)} \log p_\theta(\mathbf{x}_t | \mathbf{z}_t) \\ &\quad + f_{\beta_t \sim \mathcal{R}}(\beta_t, \mathcal{E}) D_{KL}(q_\phi(\mathbf{z}_t | \mathbf{x}_t) \| p(\mathbf{z})) \\ &= E_{\mathbf{x}_{t+1} \sim p_{\mathcal{X}}} [\mathcal{L}_R + f(\beta_t) \mathcal{L}_I + \mathcal{E}(\Delta \mathcal{L}_{\text{ELBO}})] \\ &= E_{\mathbf{x}_{t+1} \sim p_{\mathcal{X}}} [\mathcal{L}_R + f(\beta_t) \mathcal{L}_I \\ &\quad + \mathcal{E}(\Delta \mathcal{L}_{\text{ELBO}_{t+1}} + ||K L_{t+1}(\beta_{t+1}) - c||)]. \end{aligned}$$

According to the VIB theory, R_t sets the upper bound on IB $I_t(\mathbf{X}_t, \mathbf{Z}_t)$ at the t th iteration

$$\begin{aligned} R_t &= \int q(\mathbf{x}_t) d\mathbf{x} \int q_\theta(\mathbf{z}_t | \mathbf{x}_t) \log \frac{q_\theta(\mathbf{z}_t | \mathbf{x}_t)}{p(\mathbf{z}_t)} d\mathbf{z} \\ &= I_t(\mathbf{X}_t, \mathbf{Z}_t) + TC(\mathbf{Z}_t) \\ &\geq I_t(\mathbf{X}_t, \mathbf{Z}_t). \end{aligned}$$

D_t sets the lower bound

$$\begin{aligned} I_t &= \int \int p(\mathbf{x}_t, \mathbf{z}_t) \log \frac{p(\mathbf{x}_t | \mathbf{z}_t)}{p(\mathbf{x}_t)} d\mathbf{x} d\mathbf{z} \\ &\geq \int \int p(\mathbf{x}_t, \mathbf{z}_t) \log \frac{q(\mathbf{x}_t | \mathbf{z}_t)}{p(\mathbf{x}_t)} d\mathbf{x} d\mathbf{z} \\ &= \int \int p(\mathbf{x}_t, \mathbf{z}_t) \log q(\mathbf{x}_t | \mathbf{z}_t) d\mathbf{x} d\mathbf{z} \\ &\quad - \int p(\mathbf{x}_t) \log p(\mathbf{x}_t) d\mathbf{x} \\ &= \int \int p(\mathbf{x}_t, \mathbf{z}_t) \log q(\mathbf{x}_t | \mathbf{z}_t) d\mathbf{x} d\mathbf{z} + H(\mathbf{X}) \\ &= H - D_t. \end{aligned}$$

After optimized by the variational evolutionary learner $\mathcal{E}$, the loss function of eVAE constrained by a dynamic lower bound is as follows:

$$\begin{aligned} \mathcal{L}_{\text{eVAE}} &= E_{\mathbf{z} \sim q_\phi(\mathbf{z}_t | \mathbf{x}_t)} \log p_\theta(\mathbf{x}_t | \mathbf{z}_t) \\ &\quad + f_{\beta_t \sim \mathcal{R}}(\beta_t, \mathcal{E}) D_{KL}(q_\phi(\mathbf{z}_t | \mathbf{x}_t) \| p(\mathbf{z})) \\ &= E_{\mathbf{x}_{t+1} \sim p_{\mathcal{X}}} [\mathcal{L}_R + f(\beta_t) \mathcal{L}_I + \mathcal{E}(\Delta \mathcal{L}_{\text{ELBO}})] \\ &= E_{\mathbf{x}_{t+1} \sim p_{\mathcal{X}}} [\mathcal{L}_R + f(\beta_t) \mathcal{L}_I \\ &\quad + \mathcal{E}(\Delta \mathcal{L}_{\text{ELBO}_{t+1}} + ||K L_{t+1}(\beta_{t+1}) - c||)] \\ &= -D_t - \beta_t R_t - \mathcal{E}(-D_{t+1} - \beta_{t+1} R_{t+1} + D_t + c) \\ &\leq -\mathcal{E}(\beta_t)D_t - \mathcal{E}(\beta_t)I_t \end{aligned}$$

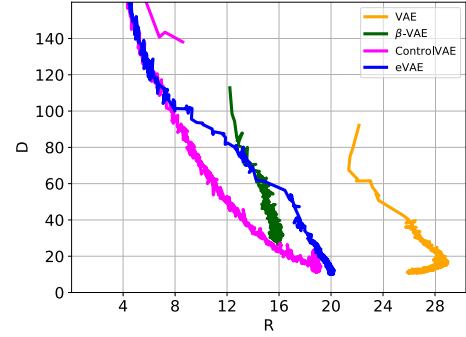


Fig. 3. Information plane with the $R - D$ curves of VAE, β -VAE, ControlVAE, and eVAE on dSprites. We plot the (R, D) values in iterations, averaged over five restarts.

where c is the desired KL point, and β_t is a chromosome sample in the t th iteration. ■

We further compare the VIB effects of eVAE against the base VAE, β -VAE, and ControlVAE per the rate-distortion ($R - D$) curve in Fig. 3. VAE (in yellow) tends to sacrifice the empirical error minimization for representation learning in the early stage. The rate is optimized to decrease at a quarter of the iterations. For β -VAE (in green), due to the experiment-specific lower bound $(-\beta_B I - D)$, the distortion cannot be minimized. ControlVAE (in pink) concentrates on initially optimizing the distortion to fit data and then optimizing the rate eventually to acquire smooth representation, resorting to the KL-specific lower bound $(-\beta_{KL} I - D)$. However, directly modifying β by the PID controller leads to the fluctuated balance between inference capacity and reconstruction quality during the iterations. Rather than monitoring the optimization process, eVAE generates an iteration-specific lower bound $(-\mathcal{E}(\beta_t)I_t - \mathcal{E}(\beta_t)D_t)$ to achieve a balance between minimizing distortion and controlling rate. In disentangled representation, the early convergence is unsuitable for reconstruction, and the unstable optimization on R harms disentanglement [33]. Only the iteration-specific lower bound-guided eVAE model can overcome the balance fluctuation problem and converge to the optimal model with a low reconstruction error and a high mutual information gap (MIG) metric.

V. EXPERIMENTS

We evaluate eVAE in three typical tasks: disentangled representation, image generation, and language modeling. The experimental settings of three learning tasks for evaluating eVAE are the same as [34]. They include the settings for the encoder-decoder network architectures and their corresponding optimizers.

A. Dataset and Baselines

Three different learning tasks and their datasets are listed below to evaluate eVAE against the VAE baselines.

1) *Disentangled Representation:* Learns independent latent variables to generate an image. We validate eVAE on dSprites,¹

¹dSprites: a disentanglement testing Sprites dataset. <https://github.com/deepmind/dsprites-dataset/>

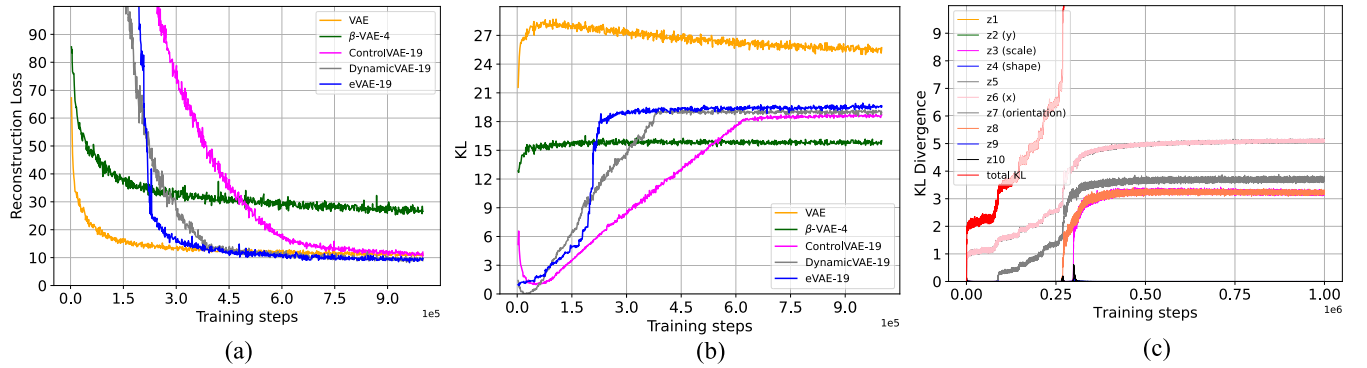


Fig. 4. Learning curves on dSprites. (a) and (b) eVAE has the lowest reconstruction loss compared with VAE ($\beta = 1$), β -VAE ($\beta = 4$), and ControlVAE ($KL = 19$) DynamicVAE ($KL = 19$) under a fixed KL point $KL = 19$. (c) Elementwise KL divergence as a function of iterations in eVAE. We can observe that eVAE retains a stable and reasonable KL divergence of each generator dimension (factor): position y (z2), scale (z3), shape (z4), position x (z6), and orientation (z7).

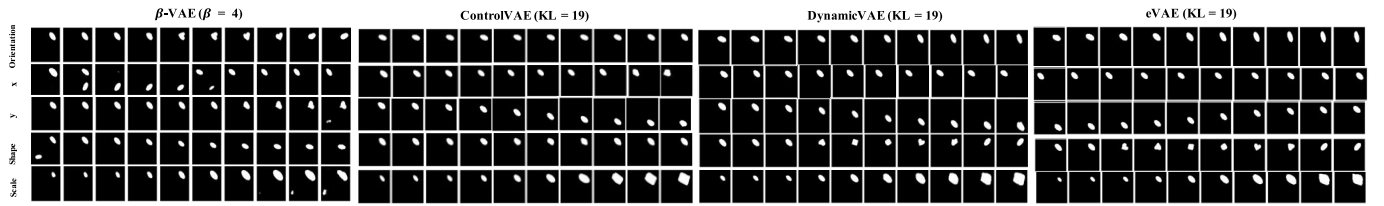


Fig. 5. Latent traversal on dSprites. To distinguish the latent factors visually, we take the ellipse for illustration. Each row represents a latent factor in the traversal while keeping others fixed. The first column refers to the seed image for initialization. The rest of the columns correspond to the images generated by manipulating the latent dimension z over its value range $[-3, 3]$.

a 2-D-shape dataset with 737 280 binary images generated by five ground-truth factors: shape (square, ellipse, heart), scale, orientation, position x , and position y . To ensure a fair and comprehensive evaluation of disentanglement performance, we compare β -VAE, ControlVAE, and DynamicVAE. For a uniform and unbiased comparison, we keep the parameters the same as in their original publications for the first two baselines, with $\beta = 1$ and $\beta = 4$. For the latter two methods, we consistently use the same setpoint, $KL = 19$.

2) *Image Generation*: Reconstructs an imagery sample based on given data points. The CelebA dataset (cropped version) [27] is used, which is a real-world celebrity face dataset with 202 599 RGB images. The baselines are β -VAE and ControlVAE.

3) *Language Modeling*: Conducts the word-level text generation. The Penn Treebank dataset (PTB) with English corpus [28] is used. To reveal the performance of restraining KL vanishing, we compare eVAE with three heuristic methods: cost annealing (noted as Cost-10k) [4], cyclical annealing (Cyc-8) [14], and PID control (PID-3) [34].

For a fair comparison, we adopt the same embedding network as β -VAE and tune eVAE to achieve the same KL set point as ControlVAE.

B. Performance of Disentangled Representation Learning

Here, we validate the performance of reconstruction with disentangled features. Fig. 4(a) shows that eVAE achieves a lower reconstruction loss of 9.2, compared with ControlVAE under the same expected KL divergence ($KL = 19$). Moreover, in Fig. 4(a) and (b), eVAE presents a more stable training curve than the other VAEs because of its dynamic weighting

TABLE I
DISENTANGLED REPRESENTATION PERFORMANCE ON DSPRITES

Metrics/Models	β -VAE ($\beta = 4$)	ControlVAE	DynamicVAE	eVAE
pos. x	0.0359	0.7697	0.7662	0.7286
pos. y	0.0243	0.7458	0.7500	0.7180
Shape	0.0116	0.0777	0.1276	0.1449
Scale	0.1507	0.6412	0.6591	0.6605
Orientation	0.0039	0.0961	0.1123	0.1261
MIG	0.1741	0.4492	0.4689	0.4723

guided by the VGA fitness function, differing from a direct modification in β -VAE and ControlVAE. The variation in each disentangled factor is illustrated in Fig. 4(c).

We show the elementwise KL divergence evolving over iterations for three VAE models: β -VAE and ControlVAE in comparison to our eVAE on dataset dSprites. We can conclude that the KL divergence of eVAE is more stable in generating elementwise disentanglement.

Fig. 5 further demonstrates the reliable disentanglement of all the factors: scale, orientation, and positions x and y , and the sharp reconstruction quality of eVAE under a set point $KL = 19$. In contrast, ControlVAE only disentangles the generative factors: scale, x and y , and DynamicVAE does not effectively disentangle on the scale, while β -VAE entangles the four factors except for scale.

Next, we use the MIG [7] and dimensionwise MIG scores to evaluate the disentanglement of different methods. The comparison performance of different methods is averaged over five random seeds. For eVAE, the disentanglement performance is better than β -VAE variants, as illustrated in Table I.

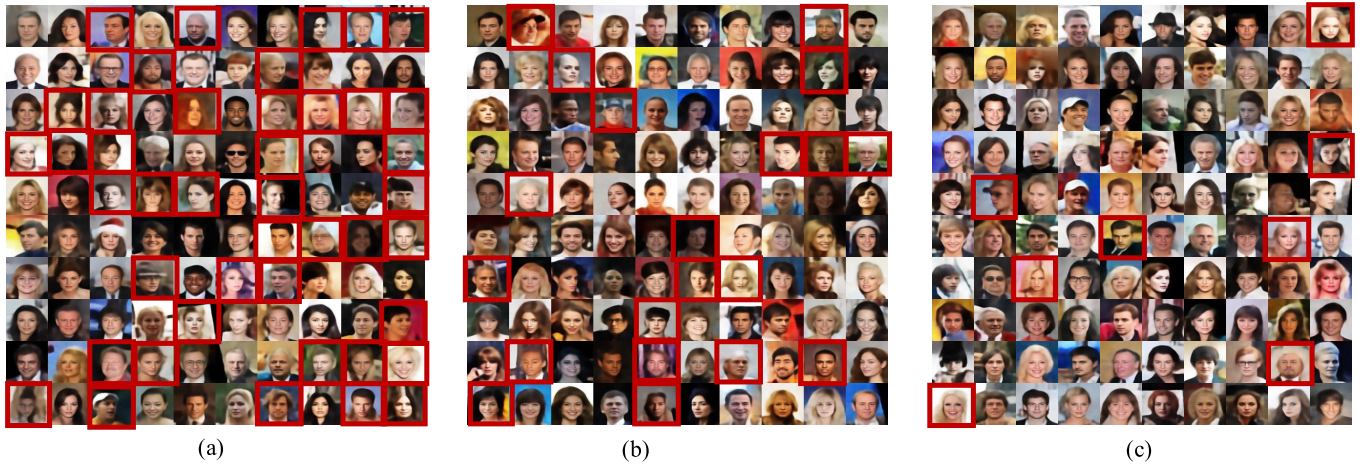


Fig. 6. Illustration of image generation on three β -VAE-based methods. In all models, images are generated by sampling from a uniform distribution on trained VAEs. Images sampled by eVAE models are blurred in line with the base model β -VAE. However, the red frames indicate that eVAE can generate more representative samples than other methods. Specifically, in (a), 38 samples marked in the red rectangle cannot generate clear human features, e.g., hair, glass, and hat, as human features are confused with the background. While (b) shows the better generation, where there are still 20 samples captured by the red frames. In contrast, the samples generated by eVAE in (c) show much better reconstruction performance with distinct foreground.

TABLE II
GENERATION PERFORMANCE ON CELEBA

Metrics/Models	VAE	ControlVAE	eVAE
FID	58.71 ± 0.207	55.79 ± 0.257	54.06 ± 0.201
SSIM	0.675 ± 0.0001	0.688 ± 0.0002	0.692 ± 0.0001

C. Performance of Image Generation

We evaluate eVAE on CelebA for image generation. For a fair and concise comparison, the KL set point is 200 for the best reconstruction quality in [34]. eVAE is tuned to achieve the desired KL point similar to ControlVAE. The learning curves can show that eVAE has the lowest reconstruction error compared with vanilla VAE and ControlVAE. Guided by the evolving β , the KL divergence in eVAE can achieve a desired point dynamically.

The test time of eVAE is to sample from the prior and generate new samples [24], and the encoder pathway is simply discarded. We adopt the same experiment setting from infoGAN [8] and β -VAE [18] for latent traverse to obtain meaningful generalization. Specifically, instead of plotting latent codes from -1 to 1 , we plot them from -3 to 3 , covering a wide region where the network is never trained. Fig. 6 illustrates the sampling figures in three models. Furthermore, we adopt the Fréchet inception distance (FID, lower better) and the distance structural similarity index (SSIM, higher better) to evaluate the generation performance. The results in Table II show that the eVAE variants improve the generation performance per both the metrics. The learning curves of image generation is in Fig. 7 and the elementwise KL divergence is shown in Fig. 8.

D. Performance of Language Modeling

To show the validity of eVAE in preventing a variational language model from degenerating into a standard one, we draw their learning curves of KL, reconstruction loss, and weight of KL over iterations.

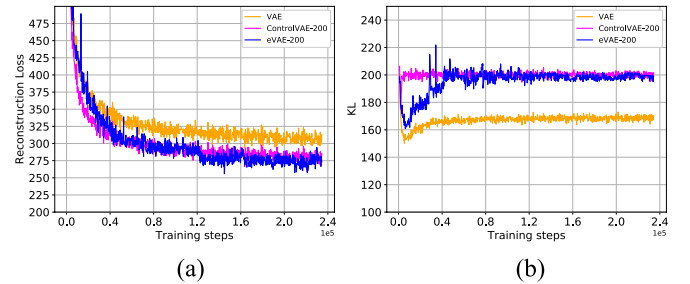


Fig. 7. Performance comparison of different VAEs for image generation on CelebA. (a) Reconstruction loss. (b) KL.

The evolution of β is shown in Fig. 9(a), demonstrating that eVAE can learn to tune β over iterations automatically rather than by a heuristic setting. Fig. 9(b) shows that the cost annealing (Cost-10k) model still suffers from KL vanishing with KL approaching 0 at the end of the iterations. In contrast, the cyclical annealing (Cyc-8), PID, and GA converge to a nonzero divergence, while eVAE achieves the lowest reconstruction loss to 70 in word generation, illustrated in Fig. 9(c). Furthermore, compared with the PID control, VGA can find a path toward a more stable stage without undergoing reconstruction fluctuations during training. We will illustrate the effects of individuals generators in Fig. 10 and crossover and mutation rates in Fig. 11.

E. Sensitivity Analysis

1) *Setpoint*: The setpoint in (15) can implicitly guide models to converge to a target value c for specific representation learning. Setpoints vary due to the difference in benchmark datasets and backbones. In this part, we show an information theory-guided principle in selecting this hyperparameter and conduct experiments on disentangled tasks to verify it.

According to Shannon's source coding theorem [16] in bit compression, a D -ary block code with blocklength n and size M has the following achievable compression limit:

$$H_D(\mathbf{X}) > \limsup_{n \rightarrow \infty} \frac{1}{n} \log_D M_n. \quad (19)$$

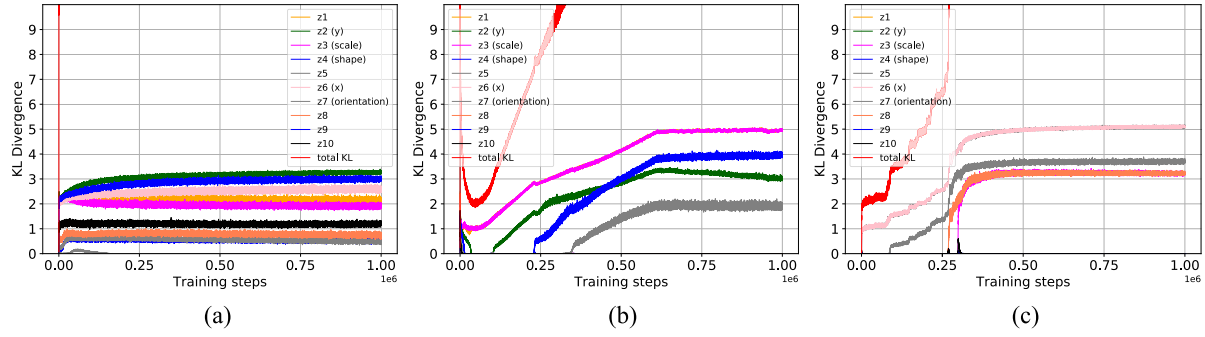


Fig. 8. Elementwise KL divergence over iterations for (a) β -VAE, (b) ControlAVE, and (c) eVAE on dSprites. It shows that each KL divergence of eVAE is larger than that of β -VAE, providing enough space for generating disentangled representation. In addition, eVAE generates more stable KL divergence compared with ControlVAE.

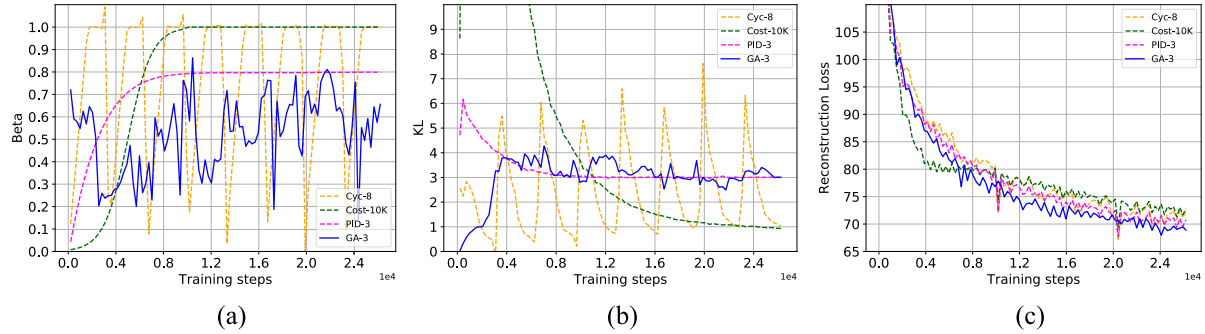


Fig. 9. Learning curves on PTB for language modeling. The suffixes of PID and VGA refer to a KL set point, i.e., $KL = 3$. Cyc-8 indicates eight iteration cycles where the weight increases linearly from 0 to 1. Cost-10k refers to 10k iterations during which the weight increases from 0 to 1 by a sigmoid function iteratively. (a) Beta. (b) KL. (c) Reconstruction loss.

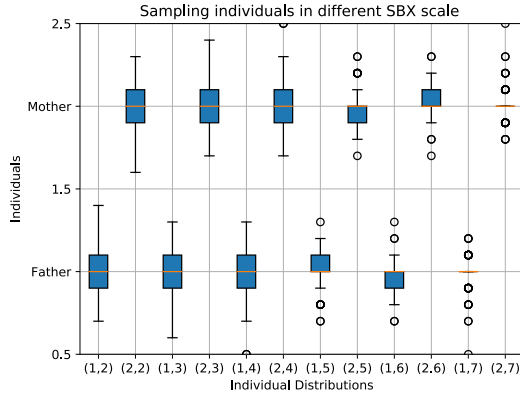


Fig. 10. Box plot of the generated distribution from the same parent (father = 1, mother = 2), varying different SBX scale η , where each offspring distribution is denoted by the tuple (child, η). We can conclude that the range of children will decrease with the growth of η . At the same time, the SBX operator will not generate diverse offspring when $\eta \geq 7$.

We can simplify the compression of VAE in binary code ($D = 2$) with dataset size L to denote the VIB c of information flow

$$c \geq H(\mathbf{X}) > \log_2 L. \quad (20)$$

Specifically, the KL setpoint should be slightly greater than $\log_2 L$. Furthermore, columns 1 and 2 in Table III illustrate that in dSprites, a large value ($30 > \log_2 737280 \approx 19$) will weaken the representation learning performance. In the CelebA dataset, the KL setpoint should be slightly greater than 190, illustrated by the results in columns 1 and 2 of Table IV for a good generation. As for the KL vanishing problem, the

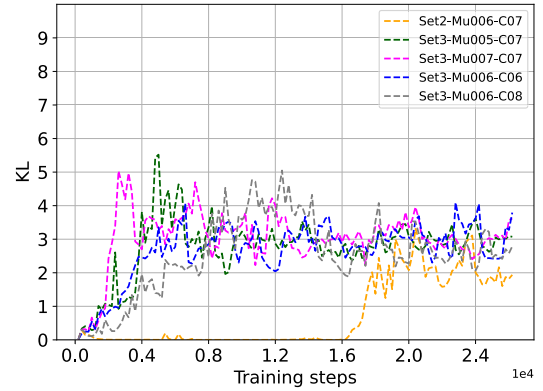


Fig. 11. Different numerical effects of the eVAE model in dealing with KL vanishing on the PTB dataset, with various setpoints, crossover rates, and mutation rates. In the different models indicated by various colors of dashed lines, the notation “Set-Mu-C,” respectively, signifies the values of the setpoint, mutation rate, and crossover rate for each model. We can deduce that an overly small setpoint (less than 3) can lead to the KL vanishing issue in the early stage of the model.

KL setpoint can be greater than 2 to avoid the KL vanishing problem, as shown in Fig. 11.

2) *Crossover and Mutation Rates*: Different from the traditional GA methods, the crossover and mutation rates in VGA should not be large to avoid convergence failure with either fluctuation or stuck to local optima. Specifically, when the crossover rate is larger than 0.5, the model cannot converge to a fixed value. On the contrary, when the individuals are crossed over with a probability less than 0.1, they tend to be the same with the strongest genes, which will damage population diversity. In that case, the inner loop will stop

TABLE III
SENSITIVITY ANALYSIS ON SETPOINTS AND CROSSOVER AND MUTATION RATES IN THE DSPRITES DATASET

Metrics/Models	eVAE					
	(KL = 30)	(KL = 15)	(0.04, 0.001)	(0.04, 0.002)	(0.03, 0.001)	(0.03, 0.002)
MIG	0.121	0.1872	0.4723	0.4715	0.4662	0.4709
Distortion	29.5	15.1	9.9	9.9	9.6	10.2

TABLE IV
SENSITIVITY ANALYSIS OF SETPOINTS AND CROSSOVER AND MUTATION RATES IN THE CELEBA DATASET

Metrics/Models	eVAE					
	(KL = 190)	(KL = 210)	(0.12, 0.005)	(0.09, 0.005)	(0.1, 0.006)	(0.1, 0.004)
FIG	58.32 $\pm$ 0.101	57.89 $\pm$ 0.205	58.07 $\pm$ 0.183	58.32 $\pm$ 0.198	57.95 $\pm$ 0.201	58.11 $\pm$ 0.189
Distortion	208	196	209	201	199	202

because there are not enough individuals to generate more genes. In addition, the mutation rate is ten times less than the crossover rate. Specifically, Table III presents the experimental configurations, as illustrated by the values in columns 3–6, for the disentanglement task. In this context, the crossover rate is recommended to be approximately 0.03, with the mutation rate being an order of magnitude less than this value. For the image generation task, columns 3–6 in Table IV show that the mutation rate must be two orders of magnitude smaller than the crossover rate. With the crossover rate around 10%, the corresponding mutation rate at approximately 0.5% yields the best generation results. For the task of language generation, as depicted in Fig. 6, the optimal crossover and mutation rates are approximately 7% and 0.6%, respectively. They can effectively prevent the KL vanishing problem.

3) *Individuals Generator*: The bitwise cross and mutation operators are unsuitable for generating diverse enough individuals for deep learning models. To generate more diversified genes, the SBX-based operator in VGA creates a wide-range scalar (float 64). Furthermore, the hyperparameter η in (11) is crucial to control the scale of sampled distribution. The box plot illustrates that a large value of η will induce a sharp distribution, where samples concentrate on their parent's values. On the contrary, a small value will generate a smooth distribution for sampling. Intuitively, η should be tuned in specific tasks to coordinate with the other two hyperparameters (mutation and crossover rates), generating the proper space to avoid unstable convergence. However, experimentally, η plays a less important role than the other two parameters. Experimentally, we set $\eta = 5$ in three representative learning tasks. Given that η directly influences the similarity between offspring and parents, theoretically, different tasks require distinct η . However, empirically, η correlates with the iteration number. By setting η to an optimal value of 5 and iterating 1 200 000 $\times$, we achieve appropriate mutation levels. Hence, η is set to 5 for all the three tasks.

VI. CONCLUSION

We make the first attempt to integrate evolutionary learning with a VAE. Accordingly, we introduce a novel evolutionary VAE (eVAE) framework. eVAE dynamically learns the

tradeoff between reconstruction effectiveness and inference robustness, which is a critical challenge facing VAEs. eVAE involves variational evolving operators to solve the premature convergence and random search problem. eVAE enables the SGD optimization and genetic algorithm synergistically by an inner–outer joint training mechanism. Following the VIB theory, eVAE derives an iteration-specific lower bound balancing the capacity of compression and decompression over iterations. Multiple task experiments show that eVAE outperforms the SOTA VAEs in traversing samples with more disentangled features, generating sharp images with lower reconstruction loss, and resolving the KL vanishing automatically in variational language modeling. We will extend eVAE to other VAE models and tasks.

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